Compute as a Commons

Mexico, Australia, and the Geography of Discovery

Jorge Luis Galvez Vallejo

National Computational Infrastructure (NCI), Australia
UNAM •
Comparative Social Dynamics: Society, Inequality, and Public Policy

Who gets to ask?

Whoever owns the computing
decides which questions get asked at all.

Rivers
Flood forecasting on the Mahanadi, with India's C-DAC

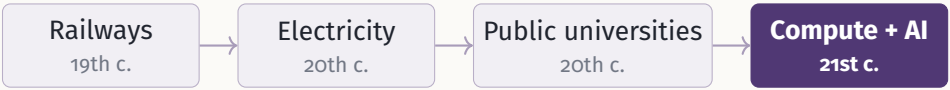
Coasts
Which shorelines get a storm-surge forecast

Crops
Which regions get a drought model

Health
Which diseases get a drug-discovery campaign

Inequality not of income, but of curiosity.

Compute is infrastructure



Build
you set the priorities

Rent
someone else sets them

Go without
your questions wait

Un bien público.

Mexico: Coatlicue

“La supercomputadora del pueblo”

314 PF
peak performance

∼14,500
GPUs

6 bn
pesos of public investment

∼7×
largest system in Latin America

- Hub of a national cluster: UNAM, IPN, Cinvestav, BUAP, UAM, and more
- Staff training at Barcelona Supercomputing Center; technical support from India’s C-DAC

Source: Secihti / ATDT announcement, Nov 2025, via Xataka México.

Australia: “very far behind”

2028

Setonix supported until; no confirmed replacement

4–5×

cost of a replacement vs. the original build

~3×

research demand vs. available capacity

“A winding down and a closure.”

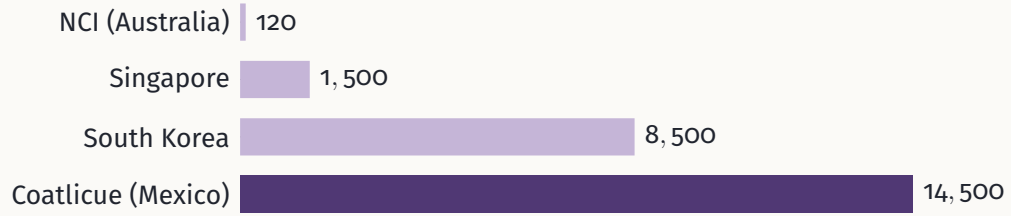
Pawsey Supercomputing Research Centre

“Very far behind” on GPUs for AI.

NCI

Source: ACS Information Age, 2026.

Access follows political decision, not national wealth



GPUs for public research. Mexico is about to have more than 100× Australia's national facility.

NCI: latest-generation GPUs. Coatlicue: planned. Sources: ACS Information Age 2026; Xataka México 2025.

A machine is not enough

People

Specialists to run it, scientists to use it, and a reason to stay when industry pays more.

Governance

Science, Pemex, tax, industry: who sets the shares? Oaxaca vs. Mexico City; regions vs. the capital.

Continuity

Machines age in 5–6 years. A permanent public service, not a one-off purchase.

A public machine with closed allocation rules just moves the queue indoors.

Compute as a commons: a Mexico–Australia proposal

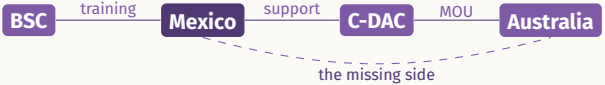
Three layers

- **Machines**, publicly owned
- **Code**, openly shared
- **People**, trained and kept

One ocean: cyclones, floods, reefs, drought, rising seas.

Three proposals

1. **Pacific modelling partnership:** joint ocean, climate and flood forecasting on both nations' public machines
2. **People exchange:** engineers and students between Coatlicue and Australia's national facilities
3. **Shared open code:** built once, adapted to the Usumacinta or the Murray–Darling



Gracias